Math 510 **Fall 2026**
Homework 1 Solutions **Drew Armstrong**

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1. Explicitly use the axioms of vector spaces to prove that

$$0\mathbf{x} = \mathbf{0} \text{ for all vectors } \mathbf{x},$$

where 0 is the zero scalar and $\mathbf{0}$ is the zero vector.

Proof. I won't bother label exactly which axioms I'm using, but here is the proof:

$$\begin{aligned} 0+0 &= 0 \\ (0+0)\mathbf{x} &= 0\mathbf{x} \\ 0\mathbf{x} + 0\mathbf{x} &= 0\mathbf{x}. \end{aligned}$$

Now subtract $0\mathbf{x}$ from both sides to obtain $0\mathbf{x} = \mathbf{0}$. $\square$

2. Let V be a vector space and consider a list of vectors $\mathbf{u}_1, \ldots, \mathbf{u}_k \in V$. Prove that the following two conditions are equivalent:

(1) For any $\mathbf{v} \in V$ there exists at most one list of scalars $a_1, \ldots, a_k$ satisfying

$$\mathbf{v} = a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k.$$

(2) For any list of scalars $a_1, \ldots, a_k$ we have

$$a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k = \mathbf{0} \quad \Longrightarrow \quad a_1 = \cdots = a_k = 0.$$

In either case we say that the vectors $\mathbf{u}_1, \ldots, \mathbf{u}_k$ are *linearly independent*.

Proof. (1)⇒(2): Assume (1) holds. To prove (2), suppose that some scalars $a_1, \ldots, a_k$ satisfy $a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k = \mathbf{0}$. Note that we also have $\mathbf{0} = 0\mathbf{u}_1 + \cdots + 0\mathbf{u}_k$.[1] But by (1), the vector $\mathbf{0}$ has at most one such expression, hence we must have $a_1 = 0, \ldots, a_k = 0$, as desired.

(2)⇒(1): Assume (2) holds. To prove (1), suppose for contradiction that there exists a vector $\mathbf{v}$ having two different expressions:

$$a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k = \mathbf{v} = b_1\mathbf{u}_1 + \cdots + b_k\mathbf{u}_k.$$

But then subtracting gives

$$(a_1 - b_1)\mathbf{u}_1 + \cdots + (a_k - b_k)\mathbf{u}_k = \mathbf{0},$$

and it follows from (2) that $a_i - b_i = 0$, hence $a_i = b_i$, for all i. This contradicts our assumption that the two expressions were different. $\square$

3. Show that the zero vector is not a member of any linearly independent set. Show that every subset of a linearly independent set is linearly independent.

Proof. Let $I = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be an independent set. To show that $\mathbf{0} \notin I$, suppose for contradiction that $\mathbf{0} \in I$. Without loss of generality we may assume that $\mathbf{v}_1 = \mathbf{0}$. But then, for example, we have

$$5\mathbf{v}_1 + 0\mathbf{v}_2 + \cdots 0\mathbf{v}_n = \mathbf{0},$$

where the first coefficient is not zero, contradicting the fact that I is independent. Next, let J be any proper subset of I. Without loss of generality we may assume that $J = \{\mathbf{v}_1, \ldots, \mathbf{v}_m\}$

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[1] It follows from Problem 1.

for some $m < n$. Suppose that $a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m = \mathbf{0}$ for some scalars $a_1, \ldots, a_m$. Let's also define the scalars b_i by $b_i = a_i$ for $i \leq m$ and $b_i = 0$ for $i > m$. Then we have

$$\begin{aligned} b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n &= a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m + 0\mathbf{v}_{m+1} + \cdots + 0\mathbf{v}_n \\ &= a_1\mathbf{v}_1 + \cdots + a_m\mathbf{v}_m \\ &= \mathbf{0}. \end{aligned}$$

Since I is independent this implies that $b_i = 0$ for all i, hence $a_i = 0$ for all i. $\square$

4. Let V be an inner product space (over $\mathbb{R}$). By definition we have $\langle \mathbf{v}, \mathbf{v} \rangle \geq 0$, hence there exists a unique non-negative square root, which we denote by

$$\|\mathbf{v}\| := \sqrt{\langle \mathbf{v}, \mathbf{v} \rangle}.$$

For any scalar $a \in \mathbb{R}$, prove that $\|a\mathbf{v}\| = |a|\|\mathbf{v}\|$.

Proof. For any vector $\mathbf{v}$ and scalar $a \in \mathbb{R}$ we have

$$\|a\mathbf{v}\| = \sqrt{\langle a\mathbf{v}, a\mathbf{v} \rangle} = \sqrt{a\langle \mathbf{v}, a\mathbf{v} \rangle} = \sqrt{a^2\langle \mathbf{v}, \mathbf{v} \rangle} = \sqrt{a^2}\sqrt{\langle \mathbf{v}, \mathbf{v} \rangle} = |a|\|\mathbf{v}\|.$$

5. Let V be an inner product space (over $\mathbb{R}$). Prove the Cauchy-Schwarz inequality:

$$|\langle \mathbf{u}, \mathbf{v} \rangle| \leq \|\mathbf{u}\|\|\mathbf{v}\|.$$

Hint: For any scalar $t \in \mathbb{R}$, show that

$$\langle \mathbf{u}, \mathbf{u} \rangle - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2\langle \mathbf{v}, \mathbf{v} \rangle = \langle \mathbf{u} - t\mathbf{v}, \mathbf{u} - t\mathbf{v} \rangle \geq 0.$$

Substitute $t = \langle \mathbf{u}, \mathbf{v} \rangle / \langle \mathbf{v}, \mathbf{v} \rangle$ into the inequality (we assume $\mathbf{v} \neq \mathbf{0}$) and then multiply both sides of the inequality by $\langle \mathbf{v}, \mathbf{v} \rangle^2$.

Proof. For any vectors $\mathbf{u}, \mathbf{v} \in V$ and scalar $t \in \mathbb{R}$ we have

$$\begin{aligned} 0 &\leq \|\mathbf{u} - t\mathbf{v}\|^2 \\ &= \langle \mathbf{u} - t\mathbf{v}, \mathbf{u} - t\mathbf{v} \rangle \\ &= \langle \mathbf{u}, \mathbf{u} \rangle - 2t\langle \mathbf{u}, \mathbf{v} \rangle + t^2\langle \mathbf{v}, \mathbf{v} \rangle. \end{aligned}$$

Since this holds for **any** scalar t we may take $t = \langle \mathbf{u}, \mathbf{v} \rangle / \langle \mathbf{v}, \mathbf{v} \rangle$.[2] Substituting this into the inequality gives

$$0 \leq \langle \mathbf{u}, \mathbf{u} \rangle - 2\frac{\langle \mathbf{u}, \mathbf{v} \rangle}{\langle \mathbf{v}, \mathbf{v} \rangle}\langle \mathbf{u}, \mathbf{v} \rangle + \frac{\langle \mathbf{u}, \mathbf{v} \rangle^2}{\langle \mathbf{v}, \mathbf{v} \rangle^2}\langle \mathbf{v}, \mathbf{v} \rangle.$$

Then multiplying both sides by the positive number $\langle \mathbf{v}, \mathbf{v} \rangle$ gives

$$\begin{aligned} 0 &\leq \langle \mathbf{u}, \mathbf{u} \rangle\langle \mathbf{v}, \mathbf{v} \rangle - 2\langle \mathbf{u}, \mathbf{v} \rangle^2 + \langle \mathbf{u}, \mathbf{v} \rangle^2 \\ 0 &\leq \langle \mathbf{u}, \mathbf{u} \rangle\langle \mathbf{v}, \mathbf{v} \rangle - \langle \mathbf{u}, \mathbf{v} \rangle^2 \\ \langle \mathbf{u}, \mathbf{v} \rangle^2 &\leq \langle \mathbf{u}, \mathbf{u} \rangle\langle \mathbf{v}, \mathbf{v} \rangle. \end{aligned}$$

Then taking square roots gives the result.

Challenge Problem (Steinitz Exchange). Let $\mathbf{u}_1, \ldots, \mathbf{u}_k$ be an independent set and let $\mathbf{v}_1, \ldots, \mathbf{v}_\ell$ be a spanning set for a vector space V. Prove that $k \leq \ell$.

[2] Here we assume that $\langle \mathbf{v}, \mathbf{v} \rangle \neq 0$ so that $\mathbf{v} \neq \mathbf{0}$. If $\mathbf{v} = \mathbf{0}$ then the Cauchy-Schwarz inequality is true because in that case $\langle \mathbf{u}, \mathbf{v} \rangle = 0$ and $\|\mathbf{u}\|\|\mathbf{v}\| = 0$ for any vector $\mathbf{u}$.

Hint: Since the v_i are spanning we can write $\mathbf{u}_1 = a_1\mathbf{v}_1 + \cdots + a_\ell \mathbf{v}_\ell$ for some a_i. Since $\mathbf{u}_1 \neq \mathbf{0}$ there exists some $a_j \neq 0$. After relabeling the $\mathbf{v}_i$ we may assume that a_1, so that

$$\mathbf{v}_1 = \frac{1}{a_1}\mathbf{u}_1 - \frac{a_2}{a_1}\mathbf{v}_2 - \cdots - \frac{a_\ell}{a_1}\mathbf{v}_\ell.$$

Prove that $\mathbf{u}_1, \mathbf{v}_2, \ldots, \mathbf{v}_\ell$ is still a spanning set, hence we can write

$$\mathbf{u}_2 = b_1\mathbf{u}_1 + b_2\mathbf{u}_2 + \cdots + b_\ell \mathbf{v}_\ell.$$

Since $\mathbf{u}_1$ and $\mathbf{u}_2$ are independent (any subset of an independent set is independent) there exists $b_j \neq 0$ with $j \geq 2$. After relabeling we may assume that $b_2 \neq 0$. Use this to show that $\mathbf{u}_1, \mathbf{u}_2, \mathbf{v}_3, \ldots, \mathbf{v}_\ell$ is still a spanning set. Continue.

Proof. (Sorry I changed the notation a bit.) Let $I = \{\mathbf{u}_1, \ldots, \mathbf{u}_m\}$ be an independent set and let $S = \{\mathbf{v}_1, \ldots, \mathbf{v}_n\}$ be a spanning set. We will show that $m \leq n$. We begin by defining $S_0 := S$. Since S_0 is a spanning set there exist scalars $a_1, \ldots, a_n \in \mathbb{R}$ such that $\mathbf{u}_1 = a_1\mathbf{v}_1 + \cdots + a_n\mathbf{v}_n$. Then since $\mathbf{u}_1 \neq \mathbf{0}$[3] there exists some index i such that $a_i \neq 0$. By relabeling the $\mathbf{v}_i$ if necessary, we may assume that $i = 1$ and hence $a_1 \neq 0$. It follows that

$$\mathbf{v}_1 = \frac{1}{a_1}\mathbf{u}_1 - \sum_{i=2}^{n} \frac{a_i}{a_1}\mathbf{v}_i.$$

Define the set $S_1 := \{\mathbf{u}_1, \mathbf{v}_2, \ldots, \mathbf{v}_n\}$ by "exchanging" $\mathbf{v}_1$ with $\mathbf{u}_1$. I claim that S_1 is still a spanning set. Indeed, consider any vector $\mathbf{w}$. Since S_0 is a spanning set we can write $\mathbf{w} = b_1\mathbf{v}_1 + \cdots + b_n\mathbf{v}_n$ for some scalars $b_1, \ldots, b_n \in \mathbb{R}$. But then we have

$$\begin{aligned}
\mathbf{w} &= b_1\left(\frac{1}{a_1}\mathbf{u}_1 - \sum_{i=2}^{n}\frac{a_i}{a_k}\mathbf{v}_i\right) + b_2\mathbf{v}_2 + \cdots + b_n\mathbf{v}_n \\
&= \frac{b_1}{a_1}\mathbf{u}_1 + \sum_{i=2}^{n}\left(b_i - \frac{a_i}{a_1}\right)\mathbf{v}_i,
\end{aligned}$$

which is in the span of S_1. Now assume for induction that we can relabel the vectors $\mathbf{v}_i$ so that $S_k = \{\mathbf{u}_1, \ldots, \mathbf{u}_k, \mathbf{v}_{k+1}, \ldots, \mathbf{v}_n\}$ is a spanning set for some $k < m$. Then we can write $\mathbf{u}_{k+1} = a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k + b_{k+1}\mathbf{v}_{k+1} + \cdots + b_n\mathbf{v}_n$ for some scalars $a_1, \ldots, a_k, b_{k+1}, \ldots, b_\ell \in \mathbb{R}$. We must have $b_i \neq 0$ for some i, since otherwise the equation $\mathbf{u}_{k+1} = a_1\mathbf{u}_1 + \cdots + a_k\mathbf{u}_k$ contradicts the independence of the $\mathbf{u}_i$. Without loss of generality we may suppose that $b_{k+1} \neq 0$, and then an argument similar to the above shows that $S_{k+1} := \{\mathbf{u}_1, \ldots, \mathbf{u}_{k+1}, \mathbf{v}_{k+2}, \ldots, \mathbf{v}_\ell\}$ is a spanning set. If $m \geq n$ then we conclude by induction that $S_n = \{\mathbf{u}_1, \ldots, \mathbf{u}_n\}$ is a spanning set. But if $m > n$ then this implies that $\mathbf{u}_{n+1} = a_1\mathbf{u}_1 + \ldots + a_n\mathbf{u}_n$ for some scalars $a_1, \ldots, a_n \in \mathbb{R}$, which contradicts the fact that the $\mathbf{u}_i$ are independent. Hence $m \leq n$. $\square$

[3] The zero vector is not an element of any independent set.